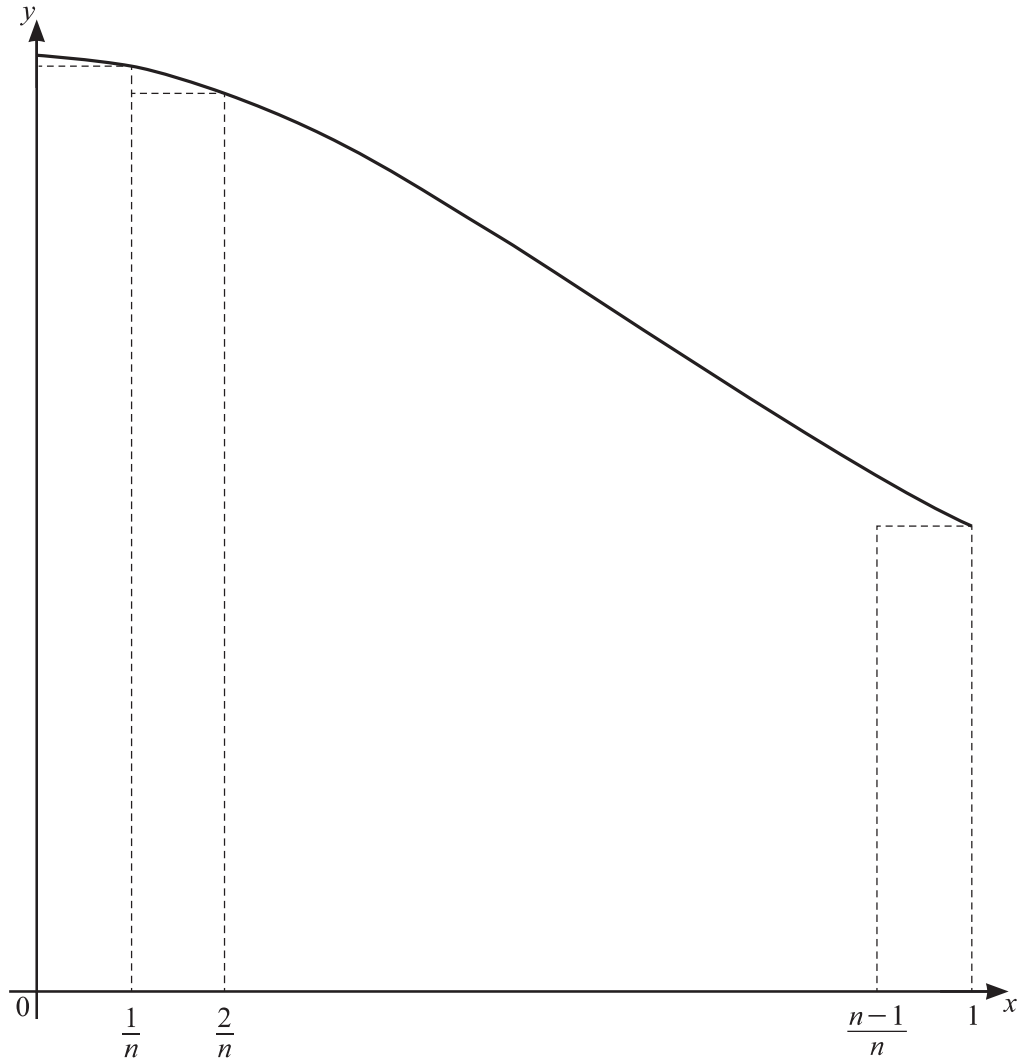


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The diagram shows the curve with equation $y = \frac{1}{x^2 + 1}$ for $0 \leq x \leq 1$, together with a set of n rectangles of width $\frac{1}{n}$.

(a) By considering the sum of the areas of these rectangles, show that

$$\sum_{r=1}^n \frac{n}{n^2 + r^2} < \frac{1}{4} \pi. \quad [5]$$

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- (b) Use a similar method to find a lower bound for $\sum_{r=1}^n \frac{n}{n^2 + r^2}$. Give your answer in terms of n and π . [4]

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- (c) Deduce the exact value of $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{n}{n^2 + r^2}$. [1]

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